

Automatic Image Analysis

Exam Preparation Questions — Answered

Parts I and II, grouped by topic

Every question from the two preparation sheets is answered below. The questions are grouped into topics; each topic opens with a short overview that ties the individual answers together, and each question keeps its original number so it can be found again on the sheet. Questions 59–61 do not appear on either sheet and are therefore missing.

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1 Foundations of Image Analysis

Q1–Q8

Big picture

Image analysis turns raw pixels (unstructured data) into statements about the scene (structured data): which objects exist, where they are, which pixel belongs to what. Processing can be driven by the data (bottom-up) or by expectations (top-down); the tasks differ in *how much* spatial detail the answer contains.

Q1. What is the difference between structured and unstructured data?

Structured data follows a predefined schema: tables, database records, sensor logs with named fields. Each value has a known meaning and can be queried directly (“all customers older than 30”). Unstructured data has no such schema: images, video, audio, free text. An image is only a grid of intensity values; the meaning (“there is a car at the left”) is implicit and must first be extracted. Automatic image analysis is exactly this conversion from unstructured to structured data.

Q2. What is top-down processing?

Model- or knowledge-driven processing. Prior knowledge, expectations, or the current task generate hypotheses (“I am looking for a face”), which then guide where and how the data is examined. Example: verifying an object hypothesis by back-projecting a model into the image, or task-driven attention.

Q3. What is bottom-up processing?

Data-driven processing. The analysis starts at the pixels and builds increasingly abstract representations without prior hypotheses: pixels → edges and features → parts → objects. Example: edge detection followed by grouping, or stimulus-driven saliency.

Q4. What is object recognition?

The umbrella term for deciding *which* objects are present in an image (and usually also where). It covers identification of a specific instance (“this is my car”) as well as categorization (“this is a car”), and contains classification and detection as special cases.

Q5. What is object detection?

Finding *all instances* of one or more object classes and localizing each of them, typically by a bounding box plus a class label and a confidence score. The number of objects is unknown in advance.

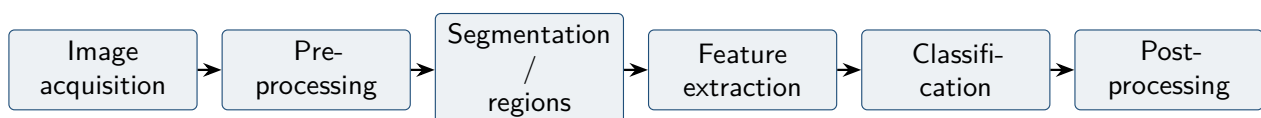
Q6. What is object classification?

Assigning a single class label to a given input: to the whole image (“this image shows a dog”) or to a given region or segment. No localization is involved.

Q7. What is semantic segmentation?

Assigning a class label to *every pixel*. The result is a dense label map. Different instances of the same class share the same label (separating them is *instance* segmentation).

Q8. How does a typical processing chain for semantic image analysis look like?



Task	Output	Spatial detail
Classification	one label per image/region	none
Detection	label + box per instance	coarse (box)
Semantic segmentation	label per pixel	dense, no instances
Instance segmentation	mask per instance	dense, with instances

Table 1: Recognition tasks ordered by the spatial detail of their output.

Acquisition delivers the image; preprocessing removes noise and normalizes contrast or geometry; segmentation (or a sliding window / region proposals) defines the units to be analyzed; feature extraction describes each unit by a vector (color, texture, HOG, SIFT, ...); a classifier maps features to labels; postprocessing adds context (non-maximum suppression, MRF/CRF smoothing, consistency checks) and produces the final interpretation. In deep networks, feature extraction and classification are learned jointly end to end.

2 Classifiers and Bag of Visual Words

Q9–Q16

Big picture

A classifier maps a feature vector $\mathbf{x}$ to a class. Linear classifiers draw one hyperplane; MLPs stack nonlinear layers to bend the decision boundary; random forests combine many randomized decision trees. Bag of Visual Words is a way to turn a variable number of local features into one fixed-length vector that any of these classifiers can take.

Q9. What are typical classifiers used in image analysis tasks?

k -nearest neighbors; Bayes / Naive Bayes classifiers (generative, based on class-conditional densities); linear classifiers (perceptron, logistic regression); support vector machines (linear or with kernels); decision trees, random forests, and boosting (AdaBoost); neural networks (MLPs and, today dominant, convolutional networks).

Q10. What is the general idea of the “Bag of (Visual) Words” approach?

Borrowed from text retrieval, where a document is represented by word counts.

1. Detect keypoints / sample patches and compute local descriptors (e.g. SIFT) on all training images.
2. Cluster the descriptors (e.g. k -means); each cluster center is a *visual word*, all centers form the *codebook* (vocabulary).
3. Represent an image by the histogram of visual-word occurrences: assign each of its descriptors to the nearest word and count.
4. Classify the fixed-length histogram (e.g. SVM, Naive Bayes).

The representation is robust to occlusion and object position but discards all spatial layout (“bag”); spatial pyramids re-add coarse layout.

Q11. What is a linear classifier?

A classifier whose decision depends on a linear function of the features,

$$g(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b, \quad \text{class} = \begin{cases} 1 & g(\mathbf{x}) > 0 \\ 2 & \text{otherwise.} \end{cases}$$

The decision boundary $g(\mathbf{x}) = 0$ is a hyperplane with normal $\mathbf{w}$ and offset b . Examples: perceptron, logistic regression, linear SVM, linear discriminant analysis. It can only separate linearly separable classes (fails on XOR), unless the features are first mapped nonlinearly.

Q12. What does MLP stand for?

Multi-Layer Perceptron.

Q13. What is the general principle of an MLP?

A feed-forward network of layers of neurons. Each neuron j computes a weighted sum of its inputs plus a bias and passes it through a nonlinear activation f :

$$a_j = \sum_i w_{ji} z_i + b_j, \quad z_j = f(a_j).$$

Input layer $\rightarrow$ one or more hidden layers $\rightarrow$ output layer, each fully connected to the next. The hidden layers learn a nonlinear feature transform so that the output layer can separate the classes; with one hidden layer of sufficient size an MLP is a universal function approximator. The weights are learned by minimizing a loss (squared error, cross-entropy) with gradient descent, gradients obtained by backpropagation.

Q14. What is usually referred to as “backprop”?

Error backpropagation: the efficient computation of the gradient of the loss E with respect to every weight via the chain rule. After a forward pass, error signals δ are propagated from the output back through the network:

$$\delta_k = f'(a_k) \frac{\partial E}{\partial z_k} (\text{output}), \quad \delta_j = f'(a_j) \sum_k w_{kj} \delta_k (\text{hidden}), \quad \frac{\partial E}{\partial w_{ji}} = \delta_j z_i.$$

The gradients are then used in a gradient-descent update (Q112). For squared error $E = \frac{1}{2}(z_k - t_k)^2$ the output term is $\delta_k = f'(a_k)(z_k - t_k)$.

Q15. What does RF stand for?

Random Forest.

Q16. What is the general principle of an RF?

An ensemble of T decision trees. Each tree routes a sample from the root to a leaf by binary tests on the features; the leaf stores a class distribution. Randomness makes the trees different (decorrelated):

- each tree is trained on a bootstrap sample of the training data (*bagging*);
- at each node only a random subset of features / candidate tests is evaluated, and the one with the highest information gain (largest entropy or Gini reduction) is chosen.

Prediction: average the leaf distributions (or majority vote) over all trees. Averaging many high-variance, low-bias trees reduces variance and overfitting. Forests are fast, handle many classes naturally, and give probabilistic outputs.

3 Fourier Transform

Q17–Q23

Big picture

The Fourier transform rewrites a signal as a sum of sinusoids. Narrow in one domain means wide in the other, convolution becomes multiplication, and a real signal has a mirror-symmetric spectrum.

Q17. What is the general principle of the Fourier transformation?

Every (sufficiently well-behaved) signal can be written as a superposition of complex exponentials $e^{i2\pi ux} = \cos(2\pi ux) + i \sin(2\pi ux)$ of different frequencies u . The transform computes, for each frequency, how much of it is in the signal (amplitude) and with which shift (phase):

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi ux} dx, \quad f(x) = \int_{-\infty}^{\infty} F(u) e^{i2\pi ux} du.$$

It is a change of basis from the spatial to the frequency domain; no information is lost. For images: 2D transform over (u, v) ; for sampled data: the DFT, computed with the FFT.

Q18. What is the Fourier spectrum of a sine function?

$$\sin(2\pi u_0 x) \longleftrightarrow \frac{i}{2} [\delta(u + u_0) - \delta(u - u_0)], \quad \cos(2\pi u_0 x) \longleftrightarrow \frac{1}{2} [\delta(u + u_0) + \delta(u - u_0)].$$

Two Dirac impulses at $\pm u_0$. For the sine they are purely imaginary and of opposite sign (odd signal); the magnitude spectrum shows two spikes of height $\frac{1}{2}$.

Q19. What is the Fourier spectrum of a rectangular function?

A sinc function. For a box of height 1 and width a :

$$\text{rect}(x/a) \longleftrightarrow a \text{sinc}(au) = \frac{\sin(\pi au)}{\pi u}.$$

Main lobe at $u = 0$ of height a , zeros at $u = k/a$. A wider box gives a narrower sinc (and vice versa).

Q20. What is the Fourier spectrum of a sinc function?

A rectangular function (duality of the transform pair): $\text{sinc}(x) \longleftrightarrow \text{rect}(u)$. This is why the sinc is the ideal low-pass filter kernel: its spectrum passes all frequencies below a cutoff and nothing above.

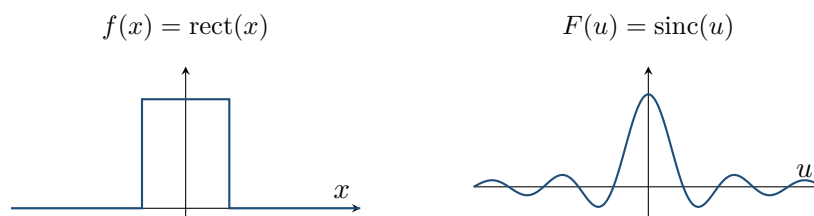


Figure 1: The rect–sinc pair. Read left to right it answers Q19, read right to left it answers Q20.

Q21. What is the convolution theorem?

Convolution in the spatial domain equals multiplication in the frequency domain, and vice versa:

$$f * g \longleftrightarrow F \cdot G, \quad f \cdot g \longleftrightarrow F * G.$$

Consequences: filtering can be done by multiplying spectra (fast for large kernels via FFT), and a filter's effect is read off its spectrum (Gaussian $\rightarrow$ Gaussian low-pass).

Q22. What is the correlation theorem?

Cross-correlation corresponds to multiplying one spectrum with the complex conjugate of the other:

$$f \star g \longleftrightarrow F^* \cdot G, \quad \text{special case: } f \star f \longleftrightarrow |F|^2.$$

The autocorrelation of a signal is the inverse transform of its power spectrum (Wiener–Khinchin). Used for fast template matching.

Q23. Which property does the spectrum of a real function fulfill?

It is Hermitian (conjugate symmetric): $F(-u) = F^*(u)$. Hence the magnitude $|F(u)|$ is even, the phase is odd, the real part is even and the imaginary part is odd. Half of the spectrum is redundant. If f is additionally even, F is real and even; if f is real and odd, F is imaginary and odd (as for the sine in Q18).

4 Fourier Descriptors

Q24–Q29

Big picture

Walk around an object's contour, write each boundary point as a complex number $z_k = x_k + i y_k$, and take the DFT. Low frequencies describe coarse shape, high frequencies fine detail. Translation, scale, rotation and start point each affect the coefficients in a simple way, so they can be normalized away.

Q24. How to extract a normalized Fourier descriptor given a grayscale image?

1. Segment the object (e.g. thresholding) to a binary image.
2. Trace the outer contour (boundary following) and sample N points, ideally equidistant along the contour.
3. Write the points as complex numbers $z_k = x_k + i y_k$, $k = 0, \dots, N - 1$.
4. Compute the DFT $Z_n = \sum_k z_k e^{-i2\pi nk/N}$.
5. Normalize: set $Z_0 = 0$ (translation), divide all coefficients by $|Z_1|$ (scale), keep only magnitudes $|Z_n|$ (rotation and start point).
6. Optionally keep only the first few low-frequency coefficients as a compact feature vector.

Q25. How does the Fourier descriptor change when the object is translated?

Only Z_0 changes; it is the (scaled) centroid of the contour. A shift $z_k \mapsto z_k + \Delta$ adds $N\Delta$ to Z_0 and leaves every Z_n , $n \neq 0$, unchanged. Normalization: drop Z_0 .

Q26. How does the Fourier descriptor change when the object is scaled?

All coefficients are multiplied by the scale factor s : $Z_n \mapsto s Z_n$ (linearity). Normalization: divide by $|Z_1|$ (or another reference coefficient).

Q27. How does the Fourier descriptor change when the object is rotated?

Rotation by φ multiplies each point by $e^{i\varphi}$, hence every coefficient: $Z_n \mapsto e^{i\varphi} Z_n$. Magnitudes stay unchanged, only phases shift. A different start point k_0 gives $Z_n \mapsto e^{i2\pi nk_0/N} Z_n$, again a pure phase change. Using $|Z_n|$ removes both.

Transformation	Effect on coefficients	Normalization
Translation by Δ	$Z_0 \mapsto Z_0 + N\Delta$, rest unchanged	set $Z_0 = 0$
Scaling by s	$Z_n \mapsto s Z_n$	divide by $ Z_1 $
Rotation by φ	$Z_n \mapsto e^{i\varphi} Z_n$	use $ Z_n $
Start point shift k_0	$Z_n \mapsto e^{i2\pi n k_0 / N} Z_n$	use $ Z_n $

Table 2: How Fourier descriptors react to similarity transforms and how each effect is removed.

Q28. How to smooth a shape using the framework of Fourier descriptors?

Low-pass filter the descriptor: keep the low-frequency coefficients Z_n for $|n| \leq M$ (with the DFT indexing, the first and the last M coefficients), set the others to zero, and apply the inverse DFT. The reconstructed contour keeps the overall shape and loses fine wiggles and noise. Fewer kept coefficients means a smoother shape (with only Z_0, Z_1 : a circle).

Q29. What are disadvantages of Fourier descriptors, i.e. when will they fail?

- They need a correct segmentation and a single closed contour; broken contours, merged objects, or noise at the boundary spoil them.
- Occlusion fails: a missing part changes all coefficients, since each coefficient is global.
- Only the outer contour is described; holes and inner structure, texture, and color are ignored; objects of several parts are hard.
- Magnitude-only descriptors discard phase, so different shapes can have the same descriptor.
- Invariance only to similarity transforms, not to perspective/affine distortion or articulation (different views of a 3D object).

5 Saliency and Scale Space

Q30–Q38

Big picture

Saliency asks *where* an image is worth looking at. The classical computational answer (Kadir/Brady) says: where the local intensity distribution is complex (high entropy) *and* this complexity is tied to a distinct scale. Evaluating “scale” needs a multi-scale representation: an image pyramid or a Gaussian scale space.

Q30. Is biological saliency a pure bottom-up, a pure top-down process, or a mixture of both?

A mixture. Bottom-up, stimulus-driven factors (contrast, color, motion, an odd item among uniform ones) attract attention automatically; top-down factors (the current task, knowledge, expectations) modulate where we look. The same scene is scanned differently when searching for different things.

Q31. What is the aim of saliency operators for image analysis?

To find regions that stand out from their surroundings, i.e. are rare, informative, and distinctive, and to restrict further (expensive) processing to them. In object models they supply repeatable interest points / regions at which descriptors are computed and matched.

Q32. Describe a computational model of saliency.

Kadir/Brady scale saliency. For each position $\mathbf{x}$ and a range of window sizes s , compute the histogram $p(I, s, \mathbf{x})$ of intensities inside the window and its entropy

$$H_D(s, \mathbf{x}) = - \sum_I p(I, s, \mathbf{x}) \log_2 p(I, s, \mathbf{x}).$$

Keep the scales s_p where H_D has a peak over s . Weight each peak by how strongly the distribution changes across scale, $W_D(s, \mathbf{x}) = s \sum_I |\partial p(I, s, \mathbf{x}) / \partial s|$. Saliency is $y(s_p, \mathbf{x}) = H_D(s_p, \mathbf{x}) W_D(s_p, \mathbf{x})$. The most salient (position, scale) pairs are selected. Noise and regular textures have high but flat entropy over scale, so they are not salient; a blob has an entropy peak where the window just encloses it.

An alternative, biologically inspired model is Itti/Koch: intensity, color and orientation channels on a Gaussian pyramid, center-surround differences, normalization, summation into one saliency map, winner-take-all with inhibition of return.

Q33. What is the entropy of a signal?

The average information content, i.e. the uncertainty of the value distribution:

$$H = - \sum_i p_i \log_2 p_i \quad [\text{bit}].$$

$H = 0$ if one value occurs with certainty; H is maximal ($\log_2 K$) for a uniform distribution over K values. For images, p_i is the normalized gray-value histogram.

Q34. Does a textured image have a higher or lower entropy than a homogeneous image?

Higher. A homogeneous image has one gray value, $p = 1$, so $H = 0$. A textured image spreads over many gray values, the histogram is flatter, so H is larger. Note: histogram entropy ignores spatial arrangement; shuffling the pixels does not change it.

Q35. What is an image pyramid?

A stack of copies of the image at decreasing resolution. Each level is obtained by low-pass filtering (Gaussian) and subsampling the previous level, typically by a factor of 2 in each direction ($R_0, R_{-1}, R_{-2}, \dots$). The total memory is at most $\frac{4}{3}$ of the original. Used for coarse-to-fine search and multi-scale detection (e.g. HOG/DPM feature pyramids). The Laplacian pyramid stores the differences between levels.

Q36. What is a linear Gaussian scale space?

A one-parameter family of images obtained by convolving the original with Gaussians of increasing width t , *without* subsampling:

$$L(\mathbf{x}, t) = L(\mathbf{x}) * G(\mathbf{x}, t), \quad G(\mathbf{x}, t) = (\pi t)^{-n/2} \exp\left(-\frac{|\mathbf{x}|^2}{t}\right).$$

All levels keep the full resolution; fine structures disappear as t grows. The Gaussian is the unique kernel for which coarser scales create no new structures (causality), and the family satisfies the diffusion (heat) equation. Difference to the pyramid: continuous scale, constant resolution, more memory.

Q37. What is the modern approach to image saliency?

Learning saliency from data with deep (convolutional) neural networks, trained on large datasets of human eye fixations or annotated salient objects, instead of hand-crafted information-theoretic measures. Trade-off: better predictions and task adaptation, but a need for much labeled data.

Q38. What are possible applications of saliency operators for images?

Interest point / region detection for matching, registration, and part-based object models (e.g. ISM); focus of attention and computation (detection candidates, robot vision); image compression (more bits for salient regions); thumbnailing, cropping, and content-aware retargeting; evaluating advertising and design; video surveillance.

6 Hough Transform

Q39–Q46

Big picture

Detect shapes by *voting*: each piece of image evidence votes for all shape parameters consistent with it; parameters that many pieces agree on appear as peaks in the accumulator. The accumulator has one dimension per unknown parameter.

Q39. State a parametric model of a line.

Slope–intercept form $y = mx + c$ (fails for vertical lines, m unbounded) or, preferred for Hough, the Hesse normal form

$$x \cos \theta + y \sin \theta = r,$$

with θ the angle of the line normal and r the distance of the line from the origin. Both parameters are bounded.

Q40. How is a point on a line in image space represented in the parameter space, if the line detector is based on the Radon transformation?

As a *sinusoid*. A single point (x_0, y_0) lies on infinitely many lines; sweeping θ gives $r(\theta) = x_0 \cos \theta + y_0 \sin \theta$. The point votes for every cell on this curve. All points of one line produce sinusoids that intersect in one common point (r, θ) : the parameters of that line.

Q41. How is a point on a line in image space represented in the parameter space, if the Hough line detector is used?

As a *single point*. The gradient-based Hough detector uses edge pixels with their gradient direction. The gradient fixes θ , and r follows from $r = x_0 \cos \theta + y_0 \sin \theta$, so each edge pixel votes for exactly one cell (r, θ) .

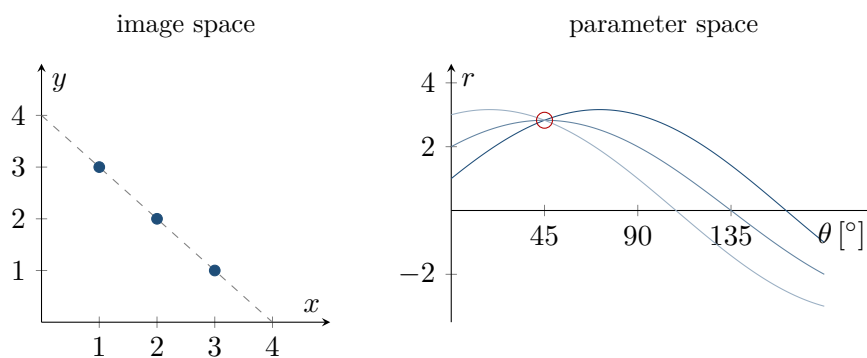


Figure 2: Radon view: the three collinear points on $x + y = 4$ trace three sinusoids that meet in one cell, $\theta = 45^\circ$, $r = 4/\sqrt{2} \approx 2.83$ (circle). With gradient information (Hough), each point would vote only for that circled cell.

Q42. What is the working principle of the Hough line detector?

1. Detect edge pixels (e.g. gradient magnitude above a threshold) and their gradient direction.
2. Discretize the parameter space (r, θ) into an accumulator $H(r, \theta)$, initialized to zero.
3. Each edge pixel votes: θ from the gradient direction, $r = x \cos \theta + y \sin \theta$, $H(r, \theta) += 1$ (or $+=$ gradient magnitude). Without gradients: vote along the full sinusoid.
4. Find local maxima in H above a threshold (non-maximum suppression); each peak is a detected line.

Robust to noise, gaps and occlusion, since missing points only lower a peak.

Q43. What is the relationship between the gradient of a line and the orientation of the line?

The image gradient at an edge is *perpendicular* to the line. Its direction $\theta = \text{atan2}(g_y, g_x)$ is exactly the normal angle θ of the Hesse form; the line itself runs at $\theta \pm 90^\circ$. In slope form, the line slope is $m = -\cot \theta = -1/\tan \theta$.

Q44. How many dimensions has a Hough circle detector?

Three: center (x_c, y_c) and radius ρ , from $(x - x_c)^2 + (y - y_c)^2 = \rho^2$. With known radius: two. Gradient direction reduces the votes per pixel: for each ρ the center lies at distance ρ along the gradient direction.

Q45. What is the R-table?

The lookup table of the Generalized Hough Transform describing an arbitrary template shape. Choose a reference point $\mathbf{a}$ (e.g. the centroid). For every boundary point $\mathbf{x}$ of the template, compute the gradient orientation ϕ and the displacement $\mathbf{r} = \mathbf{a} - \mathbf{x}$. The table is indexed by quantized ϕ ; row ϕ lists all displacement vectors of boundary points with that orientation. At detection, each edge pixel looks up its ϕ and votes for $\mathbf{x} + \mathbf{r}$ for every stored $\mathbf{r}$.

Q46. How many dimensions has the voting space of the Generalized Hough Transform?

Two (x, y of the reference point) if only translation is unknown. With unknown scale s and rotation θ : four, (x, y, s, θ) (the stored vectors are scaled and rotated before voting).

7 Part-Based Object Detection

Q47–Q58

Big picture

Instead of one rigid template, objects are modeled as a set of local parts with a spatial layout. ISM and Hough forests let parts *vote* for the object center (a learned, probabilistic Generalized Hough Transform). HOG describes a window by local gradient histograms; DPM adds deformable parts on top of HOG.

Q47. What does ISM stand for?

Implicit Shape Model (Leibe, Leonardis, Schiele).

Q48. What are the differences between the Generalized Hough Transform and the ISM?

	Generalized Hough Transform	Implicit Shape Model
Evidence	edge pixels	local appearance patches at interest points
Index	gradient orientation (R-table row)	codebook entry (visual word)
Model source	one hand-given template shape	learned from many training images
Scope	specific rigid shape	object category, intra-class variation
Votes	deterministic, equal weight	probabilistic, weighted; soft matching to several entries
Vote space	discrete accumulator	continuous (x, y, s) , modes via mean shift
Extras	—	back-projection for verification and top-down segmentation

Table 3: GHT versus ISM. The ISM keeps the voting idea but replaces the R-table by a learned codebook with occurrence distributions.

Q49. What is the algorithm to train the ISM?

1. Training images of the category with known object center and scale (bounding box, optionally figure-ground masks).
2. Detect interest points (e.g. Harris–Laplace, DoG, or salient regions) and extract patches / descriptors.
3. Cluster all descriptors (agglomerative clustering) into a codebook of visual words.
4. Match every training patch again to the codebook (all entries with similarity above a threshold t) and record, for each matching entry, the *occurrence*: relative position to the object center, scale, and optionally the local figure-ground mask.

Q50. What is the algorithm to apply the ISM?

1. Detect interest points in the test image and extract descriptors.
2. Match each descriptor to all codebook entries with similarity $> t$.
3. Each matched entry casts votes for object center and scale (x, y, s) using its stored occurrences, with weight $\frac{1}{|\text{matched entries}|} \cdot \frac{1}{|\text{occurrences}|}$.
4. Find maxima in the continuous voting space with mean-shift mode search; each mode is an object hypothesis.
5. Back-project the votes that support a hypothesis to verify it and to obtain a top-down segmentation.

Q51. What is a Hough Forest?

A random forest that performs Hough voting (Gall, Lempitsky). Trained on image patches from object and background; each leaf stores a class probability and the offset vectors from positive patches to the object center. At test time dense patches are passed through all trees, each reached leaf casts weighted votes for the object center, the votes are accumulated in a Hough image, and peaks are detections. It is a discriminatively trained codebook replacing the clustering step of the ISM.

Q52. What kind of node tests are used in a Hough Forest?

Simple binary pixel comparisons on a feature channel I^a (e.g. intensity, gradients, HOG-like channels, Lab color):

$$t(\mathcal{P}) = \begin{cases} 0 & I^a(\mathbf{p}) < I^a(\mathbf{q}) + \tau \\ 1 & \text{otherwise,} \end{cases}$$

with random channel a , positions $\mathbf{p}, \mathbf{q}$ in the patch, and threshold τ . Many random candidates are generated; the best by the energy criterion (Q53) is kept.

Q53. What kind of energy functions are used in a Hough Forest?

Two uncertainty measures, one chosen at random per node:

- *Class-label uncertainty*: entropy of the class labels in the child sets, $|A| H(A)$ summed over children (maximizing information gain).
- *Offset uncertainty*: spread of the offset vectors of the positive patches, $\sum_i \|\mathbf{d}_i - \bar{\mathbf{d}}\|^2$ per child.

The split minimizing the chosen energy is taken. Thus leaves become both pure in class and consistent in the vote they cast.

Q54. Which information is stored in the leafs of a Hough Forest?

The proportion C_L of object (foreground) patches that reached the leaf, i.e. the class probability, and the list $D_L = \{\mathbf{d}_i\}$ of offset vectors from these object patches to the object center. At test time each offset is a vote with weight $C_L/|D_L|$.

Q55. What does HOG stand for?

Histograms of Oriented Gradients (Dalal, Triggs).

Q56. How is a HOG descriptor computed given an image?

1. (Optional) gamma / color normalization.
2. Gradients with the simple filters $[-1, 0, 1]$ and $[-1, 0, 1]^T$; magnitude and orientation per pixel.
3. Divide the detection window (e.g. 64×128) into cells of 8×8 pixels. Per cell, build a histogram of 9 orientation bins over 0° – 180° ; each pixel votes with its gradient magnitude (interpolated between neighboring bins).
4. Group 2×2 cells into overlapping blocks (stride one cell) and normalize each block vector (L2-Hys) for illumination invariance.
5. Concatenate all normalized block vectors (7×15 blocks $\times 36 = 3780$ dimensions for 64×128).

Classification: linear SVM, applied with a sliding window over an image pyramid.

Q57. What does DPM stand for?

Deformable Part Model (Felzenszwalb et al.).

Q58. What is the general working principle of a DPM?

An object is a coarse *root filter* (HOG template of the whole object) plus several *part filters* at twice the resolution, each with an anchor position and a quadratic deformation cost. The score of a hypothesis at root position p_0 is

$$\text{score}(p_0) = F_0 \cdot \phi(p_0) + \sum_{i=1}^n \max_{p_i} [F_i \cdot \phi(p_i) - d_i \cdot \phi_d(\Delta p_i)] + b,$$

i.e. each part is placed where its appearance score minus deformation penalty is best. Evaluated with a sliding window over a HOG pyramid; the max over part positions is computed efficiently with generalized distance transforms. Several components (mixture) cover different viewpoints. Training: latent SVM, since part positions are not annotated, with hard-negative mining.

8 Probability and Bayesian Decisions

Q62–Q79

Big picture

All rules follow from two identities: the product rule $P(A, B) = P(A | B)P(B)$ and the sum rule $P(A) = \sum_B P(A, B)$. Bayes' theorem combines them. Classification then means picking the class with the best posterior (MAP), likelihood (ML), or lowest expected cost.

Q62. Which properties need a function to fulfill to be a probability?

Kolmogorov axioms, for events of a sample space Ω : (1) $P(A) \geq 0$; (2) $P(\Omega) = 1$; (3) additivity: for disjoint events $P(A \cup B) = P(A) + P(B)$ (countable additivity in general). Consequences: $0 \leq P(A) \leq 1$, $P(\bar{A}) = 1 - P(A)$, $P(\emptyset) = 0$.

Q63. What are the differences between a probability and a probability density?

	Probability P	Density p
Variable	discrete events	continuous variable
Range	$0 \leq P \leq 1$	$p \geq 0$, may exceed 1
Normalization	$\sum_x P(x) = 1$	$\int p(x) dx = 1$
Meaning of value	probability of the event	probability per unit (x)
Single value	$P(x)$ can be > 0	$P(X = x) = 0$; only intervals: $\int_a^b p dx$

Table 4: Probability versus probability density.

Q64. What is the joint probability of two events?

$P(A, B) = P(A \cap B)$: the probability that A and B occur together.

Q65. What is the conditional probability of an event A given B ?

The probability of A when B is known to have occurred: $P(A | B) = \frac{P(A, B)}{P(B)}$, defined for $P(B) > 0$.

Q66. What is marginalization?

Removing a variable from a joint distribution by summing (or integrating) over all its values (sum rule): $P(A) = \sum_B P(A, B) = \sum_B P(A | B)P(B)$, respectively $p(x) = \int p(x, y) dy$.

Q67. How can the joint probability of two events be factorized into probabilities of the individual events?

Product rule: $P(A, B) = P(A | B)P(B) = P(B | A)P(A)$. For many variables (chain rule): $P(x_1, \dots, x_n) = \prod_i P(x_i | x_1, \dots, x_{i-1})$.

Q68. If two events are independent, what is then known about their conditional and joint probabilities?

$P(A | B) = P(A)$, $P(B | A) = P(B)$, and $P(A, B) = P(A)P(B)$.

Q69. What is Bayes' Theorem?

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}, \quad P(B) = \sum_{A'} P(B | A')P(A').$$

It turns the probability of the data given a hypothesis into the probability of the hypothesis given the data.

Q70. What are the technical names of all terms in Bayes' theorem?

With class c and observation x : $\underbrace{P(c | x)}_{\text{posterior}} = \frac{\overbrace{P(x | c)}^{\text{likelihood}} \overbrace{P(c)}^{\text{prior}}}{\underbrace{P(x)}_{\text{evidence}}}$. The evidence is also called marginal likelihood or normalization constant.

Q71. Given the joint probability of two events, derive Bayes' theorem.

Factorize the joint in both orders (product rule): $P(A, B) = P(A | B)P(B)$ and $P(A, B) = P(B | A)P(A)$. Equate the right-hand sides, $P(A | B)P(B) = P(B | A)P(A)$, and divide by $P(B) > 0$: $P(A | B) = P(B | A)P(A)/P(B)$. The denominator follows by marginalization: $P(B) = \sum_A P(B | A)P(A)$.

Q72. What does MAP stand for?*Maximum A Posteriori.***Q73. What is the decision rule of a MAP estimator?**

Choose the class with the highest posterior:

$$\hat{c} = \arg \max_c P(c | x) = \arg \max_c P(x | c) P(c).$$

The evidence $P(x)$ is the same for all classes and can be dropped. MAP minimizes the probability of error (Bayes classifier).

Q74. What does ML stand for (in the context of probability theory)?*Maximum Likelihood.***Q75. What is the decision rule of a ML estimator?**Choose the class under which the observation is most likely, ignoring priors: $\hat{c} = \arg \max_c P(x | c)$.**Q76. In which case are ML and MAP equivalent?**

When the prior is uniform, $P(c) = \text{const}$ for all classes; it then does not change the $\arg \max$. (For parameter estimation: flat prior over parameters.)

Q77. What is Naive Bayes?

A Bayes classifier with the “naive” assumption that the features are conditionally independent given the class:

$$P(\mathbf{x} | c) = \prod_{i=1}^D P(x_i | c), \quad \hat{c} = \arg \max_c P(c) \prod_i P(x_i | c).$$

Instead of one D -dimensional density per class, only D one-dimensional densities are estimated, which beats the curse of dimensionality. It often works well even though the assumption rarely holds exactly. Zero counts are avoided by the Laplace correction (Q84).

Q78. How is it possible to take the cost of a decision into account?

Define a loss $\lambda(\alpha_i | c_j)$: the cost of action (decision) α_i when the true class is c_j . The conditional risk of an action is the expected loss under the posterior,

$$R(\alpha_i | x) = \sum_j \lambda(\alpha_i | c_j) P(c_j | x),$$

and the Bayes decision rule picks the action with minimum risk. With 0/1 loss this reduces to MAP. Asymmetric costs shift the boundary (e.g. missing a tumor costs more than a false alarm); a “reject” action with fixed cost is also possible.

Q79. What is a discriminant function?

One function $g_i(\mathbf{x})$ per class; $\mathbf{x}$ is assigned to the class with the largest value, $\hat{c} = \arg \max_i g_i(\mathbf{x})$. Decision boundaries lie where $g_i = g_j$. Choices: $g_i = P(c_i | \mathbf{x})$, $g_i = \ln p(\mathbf{x} | c_i) + \ln P(c_i)$, or $g_i = -R(\alpha_i | \mathbf{x})$; any monotonically increasing transformation gives the same decisions. For two classes a single $g = g_1 - g_2$ and its sign suffice; linear g gives a linear classifier (Q11).

9 Density Estimation

Q80–Q91

Big picture

Bayes classifiers need $p(\mathbf{x} | c)$, which must be estimated from samples. Either assume a model and fit its parameters (parametric) or let the data speak directly (nonparametric). All nonparametric estimators use $p(\mathbf{x}) \approx \frac{K}{NV}$ with K of N samples in a volume V around $\mathbf{x}$: Parzen fixes V , k -NN fixes K .

Q80. What is the difference between parametric and nonparametric methods of probability estimation?

Parametric: assume a functional form $p(\mathbf{x} | \theta)$ (e.g. Gaussian) with a fixed number of parameters and estimate θ from data. Compact and data-efficient, but biased if the assumed form is wrong. Nonparametric: no fixed form; the density is built directly from the samples. Flexible (any shape, multimodal), but needs many samples, stores the data, and suffers from the curse of dimensionality; model complexity grows with N .

Q81. What are examples for nonparametric approaches?

Histograms, Parzen windows / kernel density estimation, k -nearest neighbors.

Q82. What are examples for parametric approaches?

Gaussian (normal) distribution estimated by ML or Bayesian estimation; Bernoulli / binomial / multinomial models; exponential distribution; mixtures of Gaussians fitted by EM (semi-parametric).

Q83. What is the curse of dimensionality?

The volume of the feature space grows exponentially with the dimension D . A histogram with M bins per axis has M^D cells, so the number of samples needed to fill them grows exponentially; with a fixed dataset almost all cells are empty, data become sparse, and distances between points become similar. Nonparametric estimators (and many classifiers) therefore break down in high dimensions unless the dimension is reduced or independence is assumed (Naive Bayes).

Q84. What is the Laplace correction?

Additive smoothing of count-based probability estimates: add one pseudo-count to every value,

$$P(x = v) = \frac{n_v + 1}{N + K},$$

with n_v the count of value v , N samples and K possible values. It prevents zero probabilities for values unseen in training, which would otherwise set the whole Naive Bayes product to zero.

Q85. What is “Parzen Windows”?

A nonparametric density estimation method (kernel density estimation): the density at $\mathbf{x}$ is estimated from the samples inside a window (kernel) of fixed size h centered at $\mathbf{x}$.

Q86. How does Parzen Windows estimate a probability density?

Fix the volume $V = h^D$ and count the K samples inside: $p(\mathbf{x}) \approx \frac{K}{NV}$. Written with a kernel k ,

$$p(\mathbf{x}) = \frac{1}{N} \sum_{n=1}^N \frac{1}{h^D} k\left(\frac{\mathbf{x} - \mathbf{x}_n}{h}\right),$$

with a hypercube kernel (count) or, smoother, a Gaussian kernel. Every sample places a small bump; the estimate is their average. The bandwidth h controls smoothing: small h is spiky (overfitting), large h oversmooths.

Q87. How does k-Nearest Neighbors estimate a probability density?

Fix K and let the volume adapt: grow a sphere around $\mathbf{x}$ until it contains K samples; its volume $V(\mathbf{x})$ gives $p(\mathbf{x}) \approx \frac{K}{N V(\mathbf{x})}$. Dense regions give small volumes, sparse regions large ones. The result is not a proper density (does not integrate to 1), but for classification it yields $P(c | \mathbf{x}) = K_c/K$: majority vote among the K neighbors.

Q88. How can a Maximum-likelihood estimator be used to estimate a probability density?

Choose a parametric model $p(\mathbf{x} | \boldsymbol{\theta})$ and select the parameters under which the observed i.i.d. samples are most probable:

$$\hat{\boldsymbol{\theta}}_{ML} = \arg \max_{\boldsymbol{\theta}} \prod_{n=1}^N p(\mathbf{x}_n | \boldsymbol{\theta}) = \arg \max_{\boldsymbol{\theta}} \sum_{n=1}^N \ln p(\mathbf{x}_n | \boldsymbol{\theta}).$$

The logarithm does not change the maximizer and turns the product into a sum. Set the gradient to zero and solve; the density estimate is then $p(\mathbf{x} | \hat{\boldsymbol{\theta}}_{ML})$.

Q89. What is the ML estimate of the parameters of a Gaussian distribution?

Sample mean and sample covariance (with $1/N$):

$$\hat{\boldsymbol{\mu}} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n, \quad \hat{\boldsymbol{\Sigma}} = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \hat{\boldsymbol{\mu}})(\mathbf{x}_n - \hat{\boldsymbol{\mu}})^\top.$$

In 1D: $\hat{\sigma}^2 = \frac{1}{N} \sum (x_n - \hat{\mu})^2$. This variance estimate is biased; the unbiased one divides by $N - 1$.

Q90. What is the basic difference between a ML estimator and a Bayesian estimation of probability density?

ML treats the parameters $\boldsymbol{\theta}$ as fixed but unknown and returns a single point estimate $\hat{\boldsymbol{\theta}}$. Bayesian estimation treats $\boldsymbol{\theta}$ as a random variable with a prior $p(\boldsymbol{\theta})$, computes the posterior $p(\boldsymbol{\theta} | D) \propto p(D | \boldsymbol{\theta}) p(\boldsymbol{\theta})$, and averages over all parameters:

$$p(\mathbf{x} | D) = \int p(\mathbf{x} | \boldsymbol{\theta}) p(\boldsymbol{\theta} | D) d\boldsymbol{\theta}.$$

It includes prior knowledge and the uncertainty about the parameters.

Q91. In which case will the ML estimator and the Bayesian estimation lead to the same results?

When the posterior $p(\boldsymbol{\theta} | D)$ is sharply peaked at $\hat{\boldsymbol{\theta}}_{ML}$ (approximately a delta function). Then the integral in Q90 reduces to $p(\mathbf{x} | \hat{\boldsymbol{\theta}}_{ML})$. This happens for a large number of samples $N \rightarrow \infty$ (the likelihood dominates) when the prior is not zero at the true value, and in particular for a flat (uniform) prior, which makes the posterior mode equal to the ML estimate.

10 Graphical Models

Q92–Q107

Big picture

A graphical model draws a joint distribution as a graph: nodes are random variables, missing edges are conditional independencies. Directed graphs (Bayesian networks) factorize into conditionals; undirected graphs (MRFs) factorize into clique potentials, equivalently a sum of energies. In images, an MRF on the pixel grid combines a data term per pixel with a smoothness term between neighbors.

Q92. What are graphical models?

Probabilistic models whose joint distribution is represented by a graph. Nodes are random variables; edges express direct probabilistic dependencies; the absence of edges encodes conditional independence and thus a factorization of the joint into local factors. Types: directed (Bayesian networks), undirected (Markov / conditional random fields), factor graphs.

Q93. What are typical tasks solved with graphical models?

Inference: computing marginals $p(x_i \mid \text{evidence})$ or the most probable joint labeling (MAP), plus learning parameters and structure. In image analysis: labeling problems with context such as denoising and restoration, (semantic) segmentation, stereo matching, optical flow, and classification of image regions with neighborhood dependencies.

Q94. What is conditional independence?

A and B are conditionally independent given C , written $A \perp B \mid C$, if

$$P(A, B \mid C) = P(A \mid C) P(B \mid C) \Leftrightarrow P(A \mid B, C) = P(A \mid C).$$

Once C is known, B gives no further information about A . Example: two symptoms are dependent, but independent given the disease.

Q95. How to derive a graphical model given a factorization of the joint probability?

Directed: create one node per variable; for each factor $p(x_i \mid \text{pa}_i)$ draw an arrow from every conditioning variable in pa_i to x_i . Undirected: for each potential $\psi_C(\mathbf{x}_C)$ connect all variables in C pairwise, so C becomes a clique.

Q96. How to derive a factorization of the joint probability given a graphical model?

Directed: one factor per node, conditioned on its parents: $p(\mathbf{x}) = \prod_i p(x_i \mid \text{pa}_i)$. Undirected: one potential per maximal clique, normalized: $p(\mathbf{x}) = \frac{1}{Z} \prod_C \psi_C(\mathbf{x}_C)$.

Q97. What is a DAG?

A *Directed Acyclic Graph*: all edges are directed and there is no directed cycle. It is the graph structure of a Bayesian network; acyclicity guarantees that the product of conditionals is a valid joint distribution.

Q98. What does MRF stand for?

Markov Random Field.

Q99. What is a clique?

A subset of nodes in an undirected graph in which every pair of nodes is connected by an edge (fully connected subgraph).

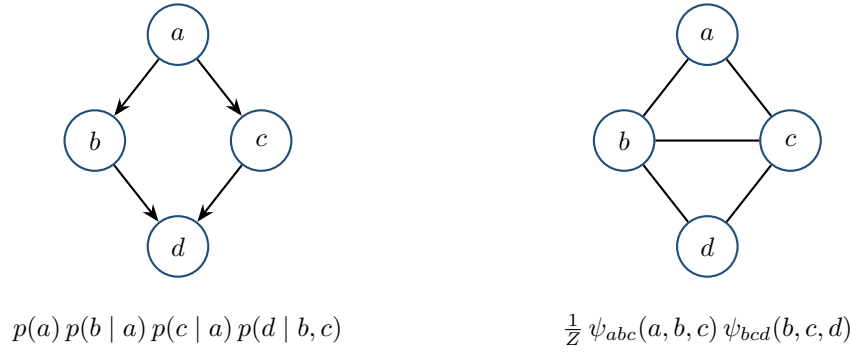


Figure 3: Left: a DAG (Bayesian network) and its factorization, one conditional per node given its parents. Right: an undirected graph (MRF) with maximal cliques $\{a, b, c\}$ and $\{b, c, d\}$ and its factorization.

Q100. What is a maximum clique?

In graphical models: a clique that cannot be extended by adding any further node without losing the clique property (strictly, a *maximal* clique; the *maximum* clique is the largest clique of the whole graph). MRF potentials are defined on these. On a 4-connected pixel grid the maximal cliques are the pairs of horizontal and vertical neighbors.

Q101. What is the general factorization for a Bayesian Network?

$$p(x_1, \dots, x_n) = \prod_{i=1}^n p(x_i | \text{pa}(x_i)),$$

with $\text{pa}(x_i)$ the parents of x_i in the DAG.

Q102. What is the general factorization for a MRF?

$$p(\mathbf{x}) = \frac{1}{Z} \prod_C \psi_C(\mathbf{x}_C), \quad Z = \sum_{\mathbf{x}} \prod_C \psi_C(\mathbf{x}_C),$$

product over the maximal cliques C , potentials $\psi_C \geq 0$ (strictly positive for the Hammersley–Clifford theorem), and partition function Z for normalization. Potentials are not probabilities.

Q103. What is the relationship between the potential and the energy of a MRF?

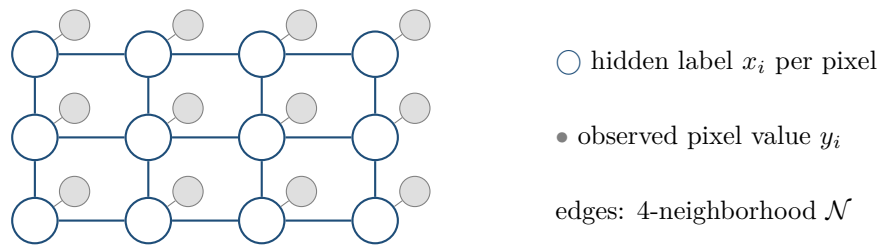
$\psi_C(\mathbf{x}_C) = \exp(-E_C(\mathbf{x}_C))$, i.e. $E_C = -\ln \psi_C$. Hence

$$p(\mathbf{x}) = \frac{1}{Z} \exp(-E(\mathbf{x})), \quad E(\mathbf{x}) = \sum_C E_C(\mathbf{x}_C)$$

(Gibbs distribution). Products of potentials become sums of energies; maximizing the probability (MAP) equals minimizing the energy.

Q104. What are unary (association) and binary / pairwise (interaction) energy terms (potentials)?

Unary (association, data) term $E_i(x_i, \mathbf{y})$: how well label x_i fits the observation at site i , usually $-\ln p(y_i | x_i)$ from a local classifier. Pairwise (interaction, smoothness) term $E_{ij}(x_i, x_j)$: cost of the label combination of neighboring sites; encodes context / prior, e.g. the Potts model $E_{ij} = \beta [x_i \neq x_j]$, which penalizes label changes between neighbors.

Q105. How does a typical MRF for image processing look like?

A grid of hidden label variables x_i (one per pixel), each connected to its observation y_i and to its 4 (or 8) neighbors. The energy is

$$E(\mathbf{x}) = \sum_i E_i(x_i, y_i) + \sum_{(i,j) \in \mathcal{N}} E_{ij}(x_i, x_j),$$

minimized (MAP labeling) with graph cuts, ICM, belief propagation, or simulated annealing. Used for denoising, segmentation, stereo.

Q106. What does CRF stand for?

Conditional Random Field.

Q107. What is the basic difference between MRF and CRF?

An MRF is *generative*: it models the joint $p(\mathbf{x}, \mathbf{y}) = p(\mathbf{y} | \mathbf{x})p(\mathbf{x})$, the prior $p(\mathbf{x})$ on labels is independent of the data, and the observations are assumed conditionally independent given the labels. A CRF is *discriminative*: it models the posterior $p(\mathbf{x} | \mathbf{y})$ directly; all potentials, including the pairwise ones, may depend on the entire observed image $\mathbf{y}$. Example: a contrast-sensitive smoothness term that penalizes label changes less across strong image edges.

11 Training Neural Networks

Q108–Q114

Big picture

An MLP is a set of weights and biases; training moves them downhill on the loss using gradients from backpropagation. Activation functions supply the nonlinearity, and momentum and weight decay make the descent faster and the result more general.

Q108. What are the weights in an MLP?

The learnable parameters on the connections: w_{ji} scales the output z_i of neuron i before it enters neuron j . Their size and sign express how strongly and in which direction an input influences a neuron. All knowledge of the trained network is stored in weights and biases.

Q109. What is the bias term in an MLP?

A constant b_j added to the weighted sum, $a_j = \sum_i w_{ji}z_i + b_j$. It shifts the activation threshold, i.e. moves the decision hyperplane away from the origin. Implemented as the weight w_{j0} of an additional input fixed to $z_0 = 1$, and learned like any other weight.

Q110. What are common activation / transfer functions in an MLP?

Function	$f(a)$	Derivative
Step (Heaviside, perceptron)	$[a \geq 0]$	0 a.e. (not usable for backprop)
Logistic sigmoid	$\sigma(a) = 1/(1 + e^{-a})$	$\sigma(1 - \sigma)$
Hyperbolic tangent	$\tanh(a)$	$1 - \tanh^2(a)$
ReLU	$\max(0, a)$	$[a > 0]$
Identity (regression output)	a	1
Softmax (class output)	$e^{a_k} / \sum_l e^{a_l}$	—

Table 5: Common activation functions.

Q111. Which properties do transfer functions in an MLP usually have?

Nonlinear (otherwise the stacked layers collapse into one linear map); differentiable, at least almost everywhere (needed for backpropagation); monotonically increasing; often bounded / saturating (sigmoid, tanh), which squashes outputs into a fixed range; derivatives cheap to compute from the function value itself.

Q112. What is the update rule for the weights in an MLP?

Gradient descent with learning rate η :

$$w_{ji}^{(t+1)} = w_{ji}^{(t)} - \eta \frac{\partial E}{\partial w_{ji}} = w_{ji}^{(t)} - \eta \delta_j z_i,$$

with δ_j from backpropagation (Q14) and z_i the input on the connection.

Q113. What is the difference between online, batch, and offline learning?

They differ in how many training samples enter one weight update:

- *Online*: update after every single sample ($\Delta w = -\eta \partial e_n / \partial w$); many cheap, noisy steps.
- *(Mini-)batch*: update after a small random subset of samples; the standard in practice (stochastic gradient descent).
- *Offline* (full batch): sum the gradient over the whole training set, then update once per epoch; exact gradient but few, expensive updates.

Note: in some texts “batch” means offline (full batch).

Q114. What is momentum / inertia and weight decay?

Momentum: add a fraction α of the previous update,

$$\Delta w^{(t)} = -\eta \frac{\partial E}{\partial w} + \alpha \Delta w^{(t-1)}, \quad 0 \leq \alpha < 1.$$

Like a heavy ball: consistent gradient directions accelerate, oscillations cancel, flat regions and small local minima are passed. *Weight decay*: add the penalty $\frac{\lambda}{2} \|\mathbf{w}\|^2$ to the loss, giving $w \leftarrow (1 - \eta\lambda)w - \eta \partial E / \partial w$. Weights shrink unless the data supports them; a regularizer against overfitting.

12 Convolutional Networks

Q115–Q120

Big picture

A ConvNet is an MLP adapted to images: neurons only see a local window (receptive field) and share the same weights across positions, so a layer is a convolution. Stacks of convolution, ReLU and pooling extract features; fully connected layers classify. Dropout, data augmentation and pre-training fight overfitting.

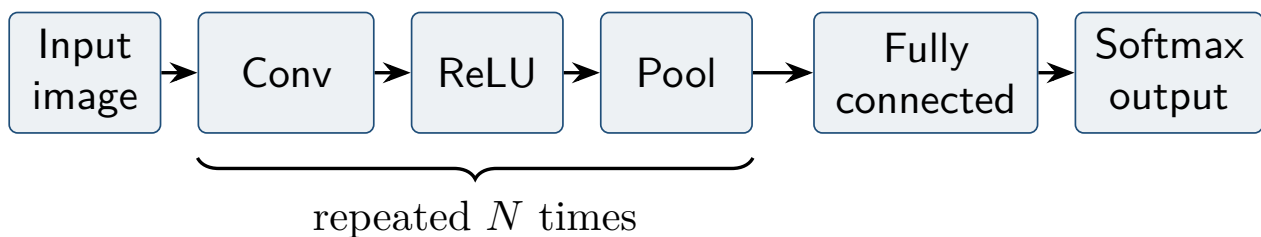
Q115. What is meant by “receptive field” and “shared weights” in the context of ConvNets?

Receptive field: the region of the input image that influences a given neuron. Each neuron connects only to a small local neighborhood (e.g. 3×3) of the previous layer; through stacking, the receptive field grows with depth. *Shared weights*: all neurons of one feature map use the same kernel weights at every position, so the layer computes a convolution. This drastically reduces the number of parameters and makes the features translation-equivariant (a pattern is detected wherever it appears).

Q116. Which activation function is usually used for ConvNets?

ReLU, $f(a) = \max(0, a)$: cheap, does not saturate for positive inputs (less vanishing gradient), sparse activations, faster training. Softmax at the output for classification.

Q117. What is the basic architecture / layout of a ConvNet?



Input $\rightarrow$ several blocks of [convolution $\rightarrow$ ReLU $\rightarrow$ (max) pooling] $\rightarrow$ one or more fully connected layers $\rightarrow$ softmax. Along the network the spatial resolution shrinks and the number of channels (feature maps) grows; features become more abstract (edges $\rightarrow$ textures $\rightarrow$ parts $\rightarrow$ objects). Examples: LeNet, AlexNet, VGG.

Q118. What is dropout?

A regularization technique: during training each neuron’s output is set to zero with probability p (e.g. 0.5), independently in every iteration. Neurons cannot rely on specific other neurons (no co-adaptation), and training effectively averages an ensemble of thinned networks. At test time all neurons are used and activations scaled by $(1 - p)$ (or, equivalently, activations are scaled by $1/(1 - p)$ during training).

Q119. What is meant with “data augmentation”?

Artificially enlarging the training set with label-preserving transformations of the existing samples: flips, random crops and translations, rotations, scaling, color and brightness jitter, noise. It reduces overfitting and teaches invariance to these transformations.

Q120. How does pre-training a ConvNet work?

Train the network first on a large dataset (e.g. ImageNet) for a generic task. Then transfer it to the target task with little data: replace the output layer to match the new classes, initialize the rest with the pre-trained weights, and *fine-tune* on the target data with a small learning rate. Alternatively freeze the early layers (generic features such as edges and textures) and train only the last layers, or use the network as a fixed feature extractor. Historically, networks were also pre-trained unsupervised, layer by layer (autoencoders, RBMs), before supervised fine-tuning.